EduBridge — Complete Project Documentation

Student Management & Hybrid Learning Platform for Classes 3–12 · Prepared for Harish Kumar D, Cloudrule Technology · Version 2.0 · 16 October 2026

1. Executive Summary

EduBridge is a hybrid ed-tech platform that connects school students (Classes 3 to 12) with teachers and institutional administrators. It replaces scattered WhatsApp groups, emails and paper homework with one secure system for study material, assignments, grading, live classes and attendance.

The platform has three deliverables:

Deliverable	Technology	Users
Student mobile app	Flutter (Android first)	Students, Classes 3–12
Admin web dashboard	React + TypeScript + Tailwind	Super Admin, Teachers
Backend API	FastAPI (Python 3.12), SQLAlchemy, PostgreSQL 16	Both clients

Supporting services: Redis (cache and queue), Celery workers, S3 or Cloudflare R2 (file storage), Firebase Cloud Messaging (push), Jitsi Meet or Zoom Video SDK (live classes), Nginx (reverse proxy and TLS), Docker Compose (deployment).

Delivery plan: Phase 1 (MVP) delivers student and admin flows with the full assignment loop. Phase 2 adds push notifications and live classes with attendance. Phase 3 adds the teacher portal, parent reports, offline caching and an AI doubt solver.

2. Problem Statement and Objectives

2.1 Problems being solved

1. **Fragmented communication.** Notes, homework and doubts are spread across WhatsApp, email and paper, so nothing is searchable or auditable.
2. **No central evaluation.** Staff cannot see who submitted what and when, and feedback is not structured or stored.

3. **Insecure live classes.** Ad-hoc meeting links get shared, strangers join, and attendance is taken by hand.
4. **Low bandwidth and low-end devices.** Many students are on 2G/3G and phones with 2–3 GB RAM. The app must load fast, cache PDFs and fail gracefully.

2.2 Objectives and success measures

Objective	Measure of success
Single source of truth for study content	100% of notes, videos and answer keys served from the platform
End-to-end assignment lifecycle	Teacher posts, student submits, teacher grades, student sees result, all inside the app
Secure live classroom	Zero shared links; join only via authenticated one-tap entry
Automatic attendance	Join time, duration and reconnects logged without manual entry
Works on weak networks	Uploads resume after drop; PDFs open offline once downloaded

3. Scope

3.1 In scope

- Authentication with phone number and password, role-based access control (RBAC).
- Class, section, subject, chapter and material management.
- Student management including bulk CSV import with generated credentials.
- Assignment publishing, multi-file student submission, grading with marks and remarks, answer-key publishing.
- Live classes: schedule, start, join, leave, attendance report (CSV).
- Push notifications through FCM.
- Audit data: last login, assignment completion rate, live attendance.

3.2 Out of scope for v1

- iOS app (Flutter code stays portable; release is a later decision).
- Online payments and fee management.
- Public student profiles or any social features.

- Proctored online exams.

4. Users, Roles and Permissions

4.1 Roles

Role	Medium	Description
STUDENT	Android app	Sees only content, assignments and live classes of their own class. Three UI modes: Classes 3–5 (large buttons, icons), Classes 6–8 (structured navigation), Classes 9–12 (dense dashboard, analytics, dark mode).
TEACHER	Web dashboard (Phase 3 scoping)	Authors material, grades work, starts live classes. Scoped to assigned classes and subjects.
SUPER_ADMIN	Web dashboard	Full control over users, classes, content, assignments, live classes and reports.

4.2 Permission matrix

Capability	STUDENT	TEACHER	SUPER_ADMIN
Log in, view own profile	Yes	Yes	Yes
View material of own class	Yes	Assigned classes	All
Upload or edit material	No	Assigned subjects	All
Create assignment	No	Assigned subjects	All
Submit assignment	Own only	No	No
Grade submission	No	Assigned subjects	All
Publish answer key	No	Assigned subjects	All
Start or schedule live class	No	Assigned classes	All
Join live class	Own class	Yes	Yes
Download attendance report	No	Assigned classes	All
Create students, classes, subjects	No	No	Yes

Capability	STUDENT	TEACHER	SUPER_ADMIN
Create teachers, change roles	No	No	Yes

Grade locking rule: every student query is filtered by the student's registered class_id on the server. A Class 4 student can never read a Class 10 resource, even with a guessed ID.

5. System Architecture

5.1 Logical components

Layer	Component	Responsibility
Client	Flutter app	Student UI, offline cache, camera scan to PDF, resumable upload, FCM receiver
Client	React admin	Content builder, student management, grading suite, live dispatcher, reports
Edge	Nginx	TLS termination, gzip, upload size limit, rate limiting, static admin build
API	FastAPI app	JWT validation, RBAC, business rules, signed file URLs
API	Modules	auth, students, content, assignments, live, notifications, reports
Async	Celery + Redis	Push fan-out, CSV import jobs, report generation, PDF stitching
Data	PostgreSQL 16	Relational data with constraints and indexes
Data	S3 / Cloudflare R2	Private bucket for PDFs, videos, submissions, answer keys
External	Firebase FCM	Push delivery to Android devices
External	Jitsi (self-hosted or 8x8) / Zoom SDK	Video rooms

5.2 Request flow

1. Client sends HTTPS request with header Authorization: Bearer <JWT>.
2. Nginx forwards to FastAPI. The auth dependency decodes the token, loads the user, checks is_active.

3. A role guard checks the endpoint's allowed roles. For students the class filter is applied automatically.
4. The module runs business logic against PostgreSQL.
5. For files, the API never returns permanent URLs. It returns signed URLs that expire in 15 minutes.
6. Slow work (push fan-out, imports) is queued to Celery and the API returns immediately.

5.3 Deployment topology

Container	Image	Port	Notes
nginx	nginx:stable	80, 443	Serves admin build, proxies /api to backend
api	edubridge-backend	8000	Uvicorn with 2–4 workers
worker	edubridge-backend	–	Celery worker, same image
db	postgres:16	5432	Volume for data, hourly backups
redis	redis:7	6379	Cache and Celery broker
jitsi (optional)	jitsi/web stack	443, 10000/udp	Only if self-hosting video

6. Functional Requirements

6.1 Authentication and profile

ID	Requirement	Priority
FR-AUTH-01	Log in with phone number and password; return JWT and profile	Must
FR-AUTH-02	Passwords stored as salted hashes (bcrypt or PBKDF2), never plain text	Must
FR-AUTH-03	Deactivated users are rejected at login and on every request	Must

ID	Requirement	Priority
FR-AUTH-04	Student can change own password; admin can reset any password	Must
FR-AUTH-05	Record last_login_at on each successful login	Should
FR-AUTH-06	OTP login by SMS as an alternative	Could (Phase 2)

6.2 Content hub

ID	Requirement	Priority
FR-CON-01	Hierarchy Subject, Chapter, Material; ordered by order_index	Must
FR-CON-02	Material types: PDF_NOTES, VIDEO_STREAM, DOC, EXTERNAL_LINK	Must
FR-CON-03	Chapters can have a release_at date; students see them only after it	Should
FR-CON-04	PDF viewer inside the app with user watermark; downloads allowed only when is_downloadable is true	Must
FR-CON-05	Video playback speed 0.75x to 2.0x; offline download with encryption	Should

6.3 Assignments

ID	Requirement	Priority
FR-ASG-01	Admin or teacher creates assignment with title, class, subject, instructions, due date, max marks, optional question file	Must
FR-ASG-02	Publishing triggers a push to all students of the class	Must
FR-ASG-03	Student submits one or more images or PDFs plus optional notes; app stitches photos into one PDF	Must
FR-ASG-04	Server enforces due date; late submission allowed only if allow_late_submission is true, then marked LATE	Must

ID	Requirement	Priority
FR-ASG-05	Student may resubmit until graded, or when status is RESUBMISSION_REQUESTED	Should
FR-ASG-06	Grader enters marks (0 to max_marks) and remarks; student is notified	Must
FR-ASG-07	Answer key stays hidden until published; publishing notifies the class	Must
FR-ASG-08	Student status shown as Pending, Submitted, Reviewed (graded) or Overdue	Must
FR-ASG-09	Submission time is the server time, never the device clock	Must

6.4 Live classes

ID	Requirement	Priority
FR-LIV-01	Start an instant class or schedule one for a class and subject	Must
FR-LIV-02	Students join with one tap; server issues a short-lived meeting token	Must
FR-LIV-03	Only students of that class can obtain a token	Must
FR-LIV-04	Attendance records join time, leave time, total seconds and reconnect count	Must
FR-LIV-05	Admin exports attendance as CSV	Must
FR-LIV-06	Students join muted; hand-raise and moderated chat available; admin can mute all	Should
FR-LIV-07	High-priority push sent when a class goes live	Must

6.5 Administration and reports

ID	Requirement	Priority
FR-ADM-01	CRUD for classes, sections, subjects, chapters, materials	Must

ID	Requirement	Priority
FR-ADM-02	Create students one by one or by CSV import; credentials generated and downloadable	Must
FR-ADM-03	Dashboard counters: students, active today, pending reviews, live now	Should
FR-ADM-04	Student activity report: last login, completion percentage, attendance percentage	Should
FR-ADM-05	Deactivate a student without deleting history	Must

7. Screen Specifications

7.1 Student mobile app

Screen	Contents and behaviour
Splash	Logo, checks stored token, routes to Login or Home
Login	Phone number, password, Login button, inline error text, no self sign-up
Home	Greeting, class label, cards for Pending assignments, Overdue count, Next live class, subject grid
Subjects	Grid of subjects for the student's class
Chapters	Ordered chapter list per subject; locked chapters show release date
Material list	Items with type icon, size, download button, offline badge
PDF viewer	Pinch zoom, page jump, watermark with student name and roll number
Video player	Speed control, resume position, download for offline
Assignments	Tabs: Pending, Submitted, Reviewed, Overdue; due date countdown
Assignment detail	Instructions, question file, answer key if published, submission status

Screen	Contents and behaviour
Submit work	Camera capture of multiple pages, reorder, crop, PDF preview, notes field, upload with progress and retry
Result	Marks out of max, remarks, link to answer key
Live classes	Today and upcoming list; LIVE badge; Join button
Live room	Jitsi/Zoom view; auto-mute; raise-hand button
Notifications	Inbox of pushes with deep links
Profile	Name, roll number, class, guardian name, change password, logout

Grade-adaptive UI: Classes 3–5 use large tiles, illustrated icons and a simplified bottom bar (Learn, Homework, Live). Classes 6–12 add search, bookmarks and dark mode. Classes 9–12 add a progress analytics card.

7.2 Admin web dashboard

Screen	Contents and behaviour
Login	Phone and password; role-based redirect
Dashboard	Counters, recent submissions, today's live classes
Classes	Table of classes and sections with student counts; create and edit
Students	Searchable table with class filter, status toggle, create, CSV import, export credentials
Student detail	Profile, activity log, assignment history, attendance
Content builder	Tree of Subject, Chapter, Material; drag and drop upload; release date picker
Assignments	List with due date and submission counts; create form with file upload
Review queue	Split view: submission files on the left, marks and remarks form on the right; next and previous navigation
Answer keys	Upload key, publish individually or in bulk after the deadline

Screen	Contents and behaviour
Live classes	Schedule form, Start now, live status, End class
Attendance	Per class report table with CSV download
Users	Teachers and admins; role assignment; password reset

8. Data Model

The PostgreSQL schema from the proposal is kept and extended. Primary keys are UUID. All timestamps are timestamptz stored in UTC.

Table	Key columns	Notes
users	id, phone_number (unique), email (unique, nullable), password_hash, full_name, role, is_active, last_login_at, created_at, updated_at	role enum SUPER_ADMIN, TEACHER, STUDENT
classes	id, grade_level (3–12), section, academic_year	unique on grade, section, year
students	id, user_id, class_id, roll_number, guardian_name, guardian_phone	unique on class_id and roll_number
subjects	id, class_id, name, code, thumbnail_url	per class
chapters	id, subject_id, title, order_index, release_at	release_at added for scheduled topics
materials	id, chapter_id, title, material_type, file_key, file_size_bytes, is_downloadable, created_at	file_key is a storage key, not a public URL
assignments	id, class_id, subject_id, title, instructions, attachment_key, answer_key_key, answer_key_published, max_marks, due_date, allow_late_submission, created_by, created_at	answer_key_published added

Table	Key columns	Notes
submissions	id, assignment_id, student_id, file_keys (JSON array), student_notes, marks_awarded, feedback_remarks, status, submitted_at, graded_at, graded_by	file_keys supports many files; unique on assignment and student
live_classes	id, class_id, subject_id, title, meeting_room_id (unique), meeting_passcode, scheduled_start, duration_minutes, status, recording_url, created_by	status SCHEDULED, IN_PROGRESS, COMPLETED, CANCELLED
live_attendance	id, live_class_id, student_id, joined_at, left_at, duration_seconds, reconnects	unique on class and student
device_tokens	id, user_id, token (unique), platform, created_at	FCM registration tokens
audit_logs	id, actor_id, action, entity, entity_id, meta (JSON), created_at	Admin actions such as grading and publishing

8.1 Relationships

- One user has zero or one student profile.
- One class has many students, subjects, assignments and live classes.
- One subject has many chapters; one chapter has many materials.
- One assignment has many submissions; each student has at most one submission per assignment.
- One live class has many attendance rows; each student has at most one row per class.

8.2 Indexes

Index	Purpose
students(class_id)	Roster queries
assignments(class_id, due_date)	Student assignment list sorted by due date
submissions(assignment_id)	Review queue

Index	Purpose
submissions(student_id)	Student history
live_classes(class_id, scheduled_start)	Upcoming class lookup
users(phone_number)	Login

8.3 Status rules

Status shown to student	Condition
Pending	No submission and now is before due date
Overdue	No submission and now is after due date
Submitted	Submission exists and status is SUBMITTED or LATE
Reviewed	Submission status is GRADED
Resubmit	Submission status is RESUBMISSION_REQUESTED

9. API Specification

9.1 Conventions

Item	Rule
Base URL	https://api.edubridge.example/api/v1
Authentication	Header Authorization: Bearer <access_token> on every endpoint except login
Content types	application/json for normal calls; multipart/form-data for uploads
Identifiers	UUID v4 strings
Timestamps	ISO 8601 in UTC, for example 2026-11-20T09:30:00Z
Pagination	Query page (default 1) and page_size (default 20, max 100)
File access	Responses carry signed URLs valid for 900 seconds; request the resource again to refresh

Paginated response shape

```
{
  "items": [],
  "page": 1,
  "page_size": 20,
  "total": 134
}
```

Error response shape (all errors)

```
{
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "marks must be between 0 and 100",
    "details": [
      { "field": "marks", "issue": "max_exceeded" }
    ]
  }
}
```

HTTP status	Meaning	Typical code
200 / 201	Success / created	none
400	Business rule violated	LATE_NOT_ALLOWED, ALREADY_GRADED
401	Missing, expired or invalid token; wrong credentials	INVALID_CREDENTIALS, TOKEN_EXPIRED
403	Role or class not permitted	FORBIDDEN
404	Resource missing or not visible to this user	NOT_FOUND
409	Duplicate	PHONE_EXISTS, ROLL_EXISTS
413	Upload larger than limit	FILE_TOO_LARGE
422	Validation failure	VALIDATION_ERROR
429	Rate limit	RATE_LIMITED
500	Server error	INTERNAL_ERROR

9.2 Endpoint index

Method	Path	Role	Purpose
POST	/auth/login	Public	Log in
GET	/auth/me	Any	Current profile
POST	/auth/change-password	Any	Change own password
POST	/devices/register	Any	Save FCM token
DELETE	/devices/{token}	Any	Remove FCM token on logout
GET	/student/dashboard	Student	Home screen data
GET	/content/subjects/{subject_id}/materials	Student, Admin	Chapter tree with signed URLs
POST	/admin/classes	Admin	Create class

Method	Path	Role	Purpose
GET	/admin/classes	Admin	List classes
POST	/admin/students	Admin	Create student
GET	/admin/students	Admin	List students
PATCH	/admin/students/{id}	Admin	Edit or deactivate
POST	/admin/students/import	Admin	Bulk CSV import
GET	/admin/students/{id}/activity	Admin	Activity report
GET	/admin/stats	Admin	Dashboard counters
POST	/admin/subjects	Admin	Create subject
POST	/admin/chapters	Admin	Create chapter
POST	/admin/materials	Admin	Upload material
DELETE	/admin/materials/{id}	Admin	Remove material
GET	/assignments/list	Student, Admin	List assignments
GET	/assignments/{id}	Student, Admin	Assignment detail
POST	/assignments/create	Admin	Create assignment
POST	/assignments/{id}/submit	Student	Submit work
GET	/assignments/{id}/my-submission	Student	Own submission and result
GET	/admin/assignments/{id}/submissions	Admin	Review queue
PATCH	/admin/submissions/{id}/grade	Admin	Grade a submission
POST	/admin/assignments/{id}/answer-key	Admin	Upload answer key
POST	/admin/assignments/{id}/publish-answer-key	Admin	Publish answer key
POST	/live/start	Admin	Start or schedule a class
GET	/live/upcoming	Student, Admin	Today and upcoming classes

Method	Path	Role	Purpose
POST	/live/{id}/join	Student, Admin	Get meeting token, log attendance
POST	/live/{id}/leave	Student	Close attendance record
POST	/live/{id}/end	Admin	End a class
GET	/live/{id}/attendance	Admin	Attendance JSON
GET	/live/{id}/attendance.csv	Admin	Attendance CSV
GET	/files/{token}	Any	Download through signed URL

9.3 Authentication

POST /auth/login

Request

```
{
  "phone": "9876543210",
  "password": "Student@123"
}
```

Response 200

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "token_type": "bearer",
  "expires_in": 43200,
  "user": {
    "id": "6f1c1b0e-6a52-4d0a-9f4b-0e2f0a1d7c11",
    "full_name": "Aarav Kumar",
    "phone_number": "9876543210",
    "role": "STUDENT",
    "student": {
      "id": "b3d6a7f2-1c3e-4a55-8a0e-2f6d9c4e1a90",
      "roll_number": "07",
      "class_id": "0d9a5c64-2b1f-47a7-9e11-3c5b8d2f6a01",
      "grade_level": 8,
      "section": "A",
      "guardian_name": "Suresh Kumar"
    }
  }
}
```

Error 401

```
{
  "error": { "code": "INVALID_CREDENTIALS", "message": "Phone or password is incorrect" }
}
```

Rules: inactive users receive 401 ACCOUNT_DISABLED. Five failed attempts from one phone within ten minutes return 429.

GET /auth/me

Returns the same user object as in the login response.

POST /auth/change-password

```
{
  "current_password": "Student@123",
  "new_password": "NewStrong#2026"
}
```

Response 200

```
{ "message": "Password updated" }
```

New password must be at least 8 characters.

9.4 Devices and push tokens

POST /devices/register

```
{
  "token": "fcm_device_registration_token_string",
  "platform": "android"
}
```

Response 200

```
{ "message": "registered" }
```

If the token already exists it is moved to the current user. Call DELETE /devices/{token} on logout.

9.5 Student dashboard

GET /student/dashboard

Response 200

```
{
  "student": {
    "full_name": "Aarav Kumar",
    "roll_number": "07",
    "grade_level": 8,
    "section": "A"
  },
  "subjects": [
    {
      "id": "2a8e5d10-77c4-4b0e-9a63-1d7f0c9b3e21",
      "name": "Science",
      "code": "SCI8",
      "thumbnail_url": null,
      "chapter_count": 12
    }
  ],
  "metrics": {
    "pending_assignments": 3,
    "overdue_assignments": 1,
    "submitted_awaiting_review": 2,
    "completion_percent": 78
  },
  "upcoming_live": [
    {
      "id": "91c0f7a3-8e2d-4d6b-b1a5-5e3c7d9f2a44",
      "title": "Light and Reflection",
      "subject_name": "Science",
      "scheduled_start": "2026-11-20T04:30:00Z",
      "duration_minutes": 45,
      "status": "SCHEDULED"
    }
  ]
}
```

9.6 Content

GET /content/subjects/{subject_id}/materials

Students get a 403 if the subject belongs to another class. Chapters with a future release_at are returned with locked set to true and no materials.

Response 200

```
{
  "subject": { "id": "2a8e5d10-77c4-4b0e-9a63-1d7f0c9b3e21", "name": "Science" },
  "chapters": [
    {
      "id": "c71b2e90-4f3a-4c1d-8d55-6a0e9b7f1c32",
      "title": "1. Crop Production",
      "order_index": 1,
      "locked": false,
      "release_at": null,
      "materials": [
        {
          "id": "e4a9d3c1-5b2f-4e7a-a8c6-0f1d2b3c4d55",
          "title": "Chapter 1 Notes",
          "material_type": "PDF_NOTES",
          "file_size_bytes": 2457600,
          "is_downloadable": true,
          "url": "https://api.edubridge.example/api/v1/files/eyJrIjoi...",
          "url_expires_at": "2026-11-20T09:45:00Z"
        },
        {
          "id": "f5b0e4d2-6c3a-4f8b-b9d7-1a2e3c4d5e66",
          "title": "Video lecture",
          "material_type": "VIDEO_STREAM",
          "file_size_bytes": 104857600,
          "is_downloadable": false,
          "url": "https://api.edubridge.example/api/v1/files/eyJrIjoi...",
          "url_expires_at": "2026-11-20T09:45:00Z"
        }
      ]
    }
  ]
}
```

9.7 Admin: classes and students

POST /admin/classes

```
{
  "grade_level": 8,
  "section": "A",
  "academic_year": "2026-27"
}
```

Response 201

```
{
  "id": "0d9a5c64-2b1f-47a7-9e11-3c5b8d2f6a01",
  "grade_level": 8,
  "section": "A",
  "academic_year": "2026-27",
  "student_count": 0
}
```

Duplicate grade, section and year returns 409 CLASS_EXISTS. grade_level outside 3 to 12 returns 422.

GET /admin/classes

Returns an array of the same class objects with student_count filled in.

POST /admin/students

```
{
  "full_name": "Aarav Kumar",
  "phone_number": "9876543210",
  "class_id": "0d9a5c64-2b1f-47a7-9e11-3c5b8d2f6a01",
  "roll_number": "07",
  "guardian_name": "Suresh Kumar",
  "guardian_phone": "9876500000",
  "password": null
}
```

If password is null the server generates a random 8 character password and returns it once.

Response 201

```
{
  "id": "b3d6a7f2-1c3e-4a55-8a0e-2f6d9c4e1a90",
  "user_id": "6f1c1b0e-6a52-4d0a-9f4b-0e2f0a1d7c11",
  "full_name": "Aarav Kumar",
  "phone_number": "9876543210",
  "class_id": "0d9a5c64-2b1f-47a7-9e11-3c5b8d2f6a01",
  "roll_number": "07",
  "generated_password": "Xk7pQ2mA"
}
```

Errors: 409 PHONE_EXISTS, 409 ROLL_EXISTS.

GET /admin/students

Query: class_id, search (name, phone or roll number), is_active, page, page_size.

```
{
  "items": [
    {
      "id": "b3d6a7f2-1c3e-4a55-8a0e-2f6d9c4e1a90",
      "full_name": "Aarav Kumar",
      "phone_number": "9876543210",
      "roll_number": "07",
      "grade_level": 8,
      "section": "A",
      "is_active": true,
      "last_login_at": "2026-11-19T14:02:11Z"
    }
  ],
  "page": 1,
  "page_size": 20,
  "total": 1
}
```

PATCH /admin/students/{id}

Any subset of: full_name, class_id, roll_number, guardian_name, guardian_phone, is_active, new_password.

```
{ "is_active": false }
```

POST /admin/students/import

Multipart field file (CSV, max 5 MB). Columns: full_name, phone_number, grade_level, section, roll_number, guardian_name, guardian_phone. The class must already exist for the current academic year.

Response 200

```
{
  "created": 48,
  "skipped": 2,
  "errors": [
    { "row": 12, "phone_number": "9876543210", "error": "PHONE_EXISTS" },
    { "row": 31, "phone_number": "98765", "error": "INVALID_PHONE" }
  ],
  "credentials_csv_url":
"https://api.edubridge.example/api/v1/files/eyJrIjoi..."
}
```

The credentials file lists phone number and generated password for every created student and expires in 15 minutes. Admin should download it immediately and distribute it to guardians.

GET /admin/students/{id}/activity

```
{
  "student_id": "b3d6a7f2-1c3e-4a55-8a0e-2f6d9c4e1a90",
  "last_login_at": "2026-11-19T14:02:11Z",
  "assignments_total": 18,
  "assignments_submitted": 14,
  "assignments_graded": 12,
  "average_percent": 81.5,
  "live_classes_total": 10,
  "live_classes_attended": 9,
  "attendance_percent": 90.0
}
```

GET /admin/stats

```
{
  "students_total": 420,
  "students_active_7d": 361,
  "submissions_pending_review": 57,
  "live_now": 2,
  "assignments_due_today": 4
}
```

9.8 Admin: subjects, chapters, materials

POST /admin/subjects

```
{
  "class_id": "0d9a5c64-2b1f-47a7-9e11-3c5b8d2f6a01",
  "name": "Science",
  "code": "SCI8"
}
```

Response 201 returns the subject object with id.

POST /admin/chapters

```
{
  "subject_id": "2a8e5d10-77c4-4b0e-9a63-1d7f0c9b3e21",
  "title": "1. Crop Production",
  "order_index": 1,
  "release_at": "2026-11-25T00:00:00Z"
}
```

POST /admin/materials

Multipart form fields:

Field	Type	Notes
chapter_id	uuid	Required
title	string	Required
material_type	enum	PDF_NOTES, VIDEO_STREAM, DOC, EXTERNAL_LINK
is_downloadable	boolean	Default true
file	file	Required unless EXTERNAL_LINK; PDF up to 25 MB, video up to 500 MB through direct-to-storage upload
external_url	string	Required only for EXTERNAL_LINK

Response 201

```
{
  "id": "e4a9d3c1-5b2f-4e7a-a8c6-0f1d2b3c4d55",
  "chapter_id": "c71b2e90-4f3a-4c1d-8d55-6a0e9b7f1c32",
  "title": "Chapter 1 Notes",
  "material_type": "PDF_NOTES",
  "file_size_bytes": 2457600,
  "is_downloadable": true
}
```

DELETE /admin/materials/{id}

Response 200 with message deleted. The storage object is removed by a background job.

9.9 Assignments

POST /assignments/create

Multipart form fields:

Field	Type	Notes
class_id	uuid	Required
subject_id	uuid	Required; must belong to class_id
title	string	Required, max 255
instructions	string	Optional
due_date	datetime	Required, ISO 8601 UTC, must be in the future
max_marks	integer	Default 100
allow_late_submission	boolean	Default false
file	file	Optional question paper (PDF or image)

Response 201

```
{
  "id": "a1b2c3d4-0001-4e5f-8a9b-1c2d3e4f5a6b",
  "class_id": "0d9a5c64-2b1f-47a7-9e11-3c5b8d2f6a01",
  "subject_id": "2a8e5d10-77c4-4b0e-9a63-1d7f0c9b3e21",
  "title": "Science Worksheet 4",
  "instructions": "Answer all questions in your notebook and upload clear
photos.",
  "due_date": "2026-11-27T18:29:00Z",
  "max_marks": 25,
  "allow_late_submission": false,
  "attachment_url": "https://api.edubridge.example/api/v1/files/eyJrIjoi...",
  "notified_students": 42
}
```

GET /assignments/list

Query: class_id (admin only), subject_id, status (PENDING, SUBMITTED, GRADED, OVERDUE), page, page_size. For students class_id is ignored and the server uses their own class.

Student response 200

```
{
  "items": [
    {
      "id": "a1b2c3d4-0001-4e5f-8a9b-1c2d3e4f5a6b",
      "title": "Science Worksheet 4",
      "subject_name": "Science",
      "due_date": "2026-11-27T18:29:00Z",
      "max_marks": 25,
      "status": "PENDING",
      "marks_awarded": null
    },
    {
      "id": "a1b2c3d4-0002-4e5f-8a9b-1c2d3e4f5a6b",
      "title": "Maths Chapter 3 Exercise",
      "subject_name": "Maths",
      "due_date": "2026-11-18T18:29:00Z",
      "max_marks": 20,
      "status": "GRADED",
      "marks_awarded": 18
    }
  ],
  "page": 1,
  "page_size": 20,
  "total": 2
}
```

Admin response items add submitted_count, graded_count and total_students.

GET /assignments/{id}

```
{
  "id": "a1b2c3d4-0001-4e5f-8a9b-1c2d3e4f5a6b",
  "title": "Science Worksheet 4",
  "subject_name": "Science",
  "instructions": "Answer all questions in your notebook and upload clear photos.",
  "due_date": "2026-11-27T18:29:00Z",
  "max_marks": 25,
  "allow_late_submission": false,
  "attachment_url": "https://api.edubridge.example/api/v1/files/eyJrIjoi...",
  "answer_key_url": null,
  "status": "PENDING"
}
```

answer_key_url is null until the key is published.

POST /assignments/{id}/submit

Multipart form fields: files (one or more, PDF or JPEG or PNG, up to 10 files, 25 MB each), notes (optional string).

Response 201

```
{
  "id": "5e6f7a8b-0001-4c2d-9e3f-4a5b6c7d8e9f",
  "assignment_id": "a1b2c3d4-0001-4e5f-8a9b-1c2d3e4f5a6b",
  "status": "SUBMITTED",
  "submitted_at": "2026-11-26T15:12:40Z",
  "files": [
    { "name": "page1.pdf", "url": "https://api.edubridge.example/api/v1/files/eyJrIjoi..." }
  ]
}
```

Business rules

Situation	Result
Before due date	status SUBMITTED
After due date and allow_late_submission true	status LATE
After due date and allow_late_submission false	400 LATE_NOT_ALLOWED
Already GRADED	400 ALREADY_GRADED
Already submitted, not graded, or status RESUBMISSION_REQUESTED	Files are replaced and submitted_at updated
Assignment belongs to another class	404 NOT_FOUND

GET /assignments/{id}/my-submission

```
{
  "id": "5e6f7a8b-0001-4c2d-9e3f-4a5b6c7d8e9f",
  "status": "GRADED",
  "submitted_at": "2026-11-26T15:12:40Z",
  "marks_awarded": 22,
  "max_marks": 25,
  "feedback_remarks": "Good diagrams. Revise the steps of irrigation.",
  "graded_at": "2026-11-28T06:10:00Z",
  "files": [
    { "name": "page1.pdf", "url":
"https://api.edubridge.example/api/v1/files/eyJrIjoi..." }
  ],
  "answer_key_url": "https://api.edubridge.example/api/v1/files/eyJrIjoi..."
}
```

9.10 Grading and answer keys

GET /admin/assignments/{id}/submissions

Query: status, search, page, page_size. Students who have not submitted are included with status PENDING so staff can see the full roster.

```
{
  "assignment": { "id": "a1b2c3d4-0001-4e5f-8a9b-1c2d3e4f5a6b", "title":
"Science Worksheet 4", "max_marks": 25 },
  "items": [
    {
      "submission_id": "5e6f7a8b-0001-4c2d-9e3f-4a5b6c7d8e9f",
      "student_id": "b3d6a7f2-1c3e-4a55-8a0e-2f6d9c4e1a90",
      "student_name": "Aarav Kumar",
      "roll_number": "07",
      "status": "SUBMITTED",
      "submitted_at": "2026-11-26T15:12:40Z",
      "files": [ { "name": "page1.pdf", "url":
"https://api.edubridge.example/api/v1/files/eyJrIjoi..." } ],
      "student_notes": "Question 5 is on the last page"
    },
    {
      "submission_id": null,
      "student_id": "c4e7b8a3-2d4f-4b66-9b1f-3a7e0d5f2b01",
      "student_name": "Diya Sharma",
      "roll_number": "08",
      "status": "PENDING",
      "submitted_at": null,
      "files": []
    }
  ],
  "page": 1,
  "page_size": 50,
  "total": 42
}
```

PATCH /admin/submissions/{id}/grade

Request

```
{
  "marks": 22,
  "remarks": "Good diagrams. Revise the steps of irrigation.",
  "request_resubmission": false
}
```

Response 200

```
{
  "id": "5e6f7a8b-0001-4c2d-9e3f-4a5b6c7d8e9f",
  "status": "GRADED",
  "marks_awarded": 22,
  "feedback_remarks": "Good diagrams. Revise the steps of irrigation.",
  "graded_at": "2026-11-28T06:10:00Z",
  "graded_by": "1a2b3c4d-0000-4a5b-8c9d-0e1f2a3b4c5d"
}
```

Rules: marks must be between 0 and the assignment max_marks, otherwise 422. If request_resubmission is true the status becomes RESUBMISSION_REQUESTED, marks are optional and the student is notified. A push is sent to the student after grading.

POST /admin/assignments/{id}/answer-key

Multipart field file (PDF or image). Stores the key but keeps it hidden.

```
{ "message": "Answer key uploaded", "published": false }
```

POST /admin/assignments/{id}/publish-answer-key

Request body is optional. Publishing is refused before the due date unless force is true.

```
{ "force": false }
```

Response 200

```
{ "message": "Answer key published", "published": true, "notified_students": 42 }
```

9.11 Live classes

POST /live/start

Request. Leave scheduled_start null to start immediately.

```
{
  "class_id": "0d9a5c64-2b1f-47a7-9e11-3c5b8d2f6a01",
  "subject_id": "2a8e5d10-77c4-4b0e-9a63-1d7f0c9b3e21",
  "title": "Light and Reflection",
  "scheduled_start": "2026-11-20T04:30:00Z",
  "duration_minutes": 45
}
```

Response 201

```
{
  "id": "91c0f7a3-8e2d-4d6b-b1a5-5e3c7d9f2a44",
  "title": "Light and Reflection",
  "status": "SCHEDULED",
  "scheduled_start": "2026-11-20T04:30:00Z",
  "duration_minutes": 45,
  "meeting_room_id": "eb-8f3a1c9d2e7b4a60",
  "notified_students": 42
}
```

When started immediately, status is IN_PROGRESS and the push uses high priority.

GET /live/upcoming

Students see classes of their own class from the start of today onward, ordered by scheduled_start. Query: include_completed (default false).

```
{
  "items": [
    {
      "id": "91c0f7a3-8e2d-4d6b-b1a5-5e3c7d9f2a44",
      "title": "Light and Reflection",
      "subject_name": "Science",
      "scheduled_start": "2026-11-20T04:30:00Z",
      "duration_minutes": 45,
      "status": "IN_PROGRESS",
      "can_join": true
    }
  ]
}
```

can_join is true from 10 minutes before the start until the class ends. The meeting token is not included here; it is issued by join.

POST /live/{id}/join

Creates or updates the attendance row. A second join by the same student increments reconnects.

```
{
  "provider": "jitsi",
  "domain": "meet.eduBridge.example",
  "room": "eb-8f3a1c9d2e7b4a60",
  "jwt": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "display_name": "Aarav Kumar",
  "start_muted": true,
  "is_moderator": false,
  "expires_in": 3600
}
```

Errors: 403 NOT_YOUR_CLASS, 400 CLASS_NOT_STARTED, 400 CLASS_ENDED.

POST /live/{id}/leave

```
{ "message": "left", "duration_seconds": 2412 }
```

Duration accumulates across reconnects. If the app is killed without calling leave, the server closes open rows when the class ends, using the end time.

POST /live/{id}/end

```
{ "status": "COMPLETED", "attendees": 39 }
```

GET /live/{id}/attendance

```
{
  "live_class": { "id": "91c0f7a3-8e2d-4d6b-b1a5-5e3c7d9f2a44", "title": "Light and Reflection", "duration_minutes": 45 },
  "summary": { "enrolled": 42, "attended": 39, "absent": 3, "average_minutes": 38 },
  "items": [
    {
      "student_id": "b3d6a7f2-1c3e-4a55-8a0e-2f6d9c4e1a90",
      "student_name": "Aarav Kumar",
      "roll_number": "07",
      "joined_at": "2026-11-20T04:28:10Z",
      "left_at": "2026-11-20T05:08:22Z",
      "duration_seconds": 2412,
      "reconnects": 1,
      "present": true
    }
  ]
}
```

GET /live/{id}/attendance.csv

Content-Type text/csv. Columns: roll_number, student_name, present, joined_at, left_at, duration_minutes, reconnects.

9.12 File download

GET /files/{token}

The token is an HMAC-signed value that encodes the storage key and expiry time. The server checks the signature and expiry, blocks path traversal, then streams the file with the right Content-Type. Expired or tampered tokens return 403 LINK_EXPIRED. When storage is S3 or R2, the API returns a native presigned URL with the same 15 minute lifetime instead.

10. Push Notifications (FCM)

10.1 Events

Event	Recipients	Priority	Title and body	Deep link
ASSIGNMENT_CREATED	All active students of the class	Normal	New Science assignment: Worksheet 4. Due Fri 27 Nov	/assignments/{id}
ASSIGNMENT_DUE_SOON	Students with no submission	Normal	Worksheet 4 is due in 24 hours	/assignments/{id}
SUBMISSION_GRADED	The one student	Normal	Your Science work is graded: 22/25	/assignments/{id}/result
RESUBMISSION_REQUESTED	The one student	Normal	Please resubmit Science Worksheet 4	/assignments/{id}
ANSWER_KEY_PUBLISHED	All students of the class	Normal	Answer key available for Worksheet 4	/assignments/{id}
LIVE_STARTED	All students of the class	High	Science class is live now. Tap to join	/live/{id}
LIVE_SCHEDULED	All students of the class	Normal	Science class on Fri at 10:00 AM	/live/{id}
LIVE_REMINDER	All students of the class	High	Class starts in 10 minutes	/live/{id}

Event	Recipients	Priority	Title and body	Deep link
MATERIAL_ADDED	All students of the class	Normal	New notes added in Science: Crop Production	/subjects/{id}

10.2 Payload sent to FCM

```
{
  "message": {
    "token": "fcm_device_registration_token_string",
    "notification": {
      "title": "Science class is live now",
      "body": "Light and Reflection. Tap to join"
    },
    "data": {
      "event": "LIVE_STARTED",
      "entity_id": "91c0f7a3-8e2d-4d6b-b1a5-5e3c7d9f2a44",
      "deep_link": "/live/91c0f7a3-8e2d-4d6b-b1a5-5e3c7d9f2a44"
    },
    "android": {
      "priority": "high",
      "notification": { "channel_id": "live_classes", "sound": "default" }
    }
  }
}
```

10.3 Delivery rules

- Fan-out runs in a Celery task. The API call that triggers it returns without waiting.
- Tokens are sent in batches of 500. Tokens FCM reports as unregistered are deleted from device_tokens.
- A user with several devices receives the push on every device.
- Each push is also stored in an in-app notifications list so a missed push can still be read.
- Android channels: live_classes (high importance), homework (default), general (low).

11. Live Class Integration

11.1 Provider choice

Option	Cost	Control	Notes
Self-hosted Jitsi Meet	Server cost only	Full; JWT auth, moderator rules, branding	Recommended; needs a server with UDP port 10000 open
Jitsi as a Service (8x8)	Per monthly active user	JWT auth supported	Fastest to launch, no server work
Zoom Video SDK	Per minute	Strong, with recording	Higher cost, more native-SDK work
Public meet.jit.si	Free	No custom JWT	Not suitable: rooms cannot be locked to a class

The design below uses Jitsi with JWT. The same join endpoint can return a Zoom signature instead by changing the provider field.

11.2 Token (JWT) claims issued by POST /live/{id}/join

```
{
  "aud": "jitsi",
  "iss": "edubridge-app-id",
  "sub": "meet.eduBridge.example",
  "room": "eb-8f3a1c9d2e7b4a60",
  "exp": 1795233600,
  "nbf": 1795230000,
  "context": {
    "user": {
      "id": "b3d6a7f2-1c3e-4a55-8a0e-2f6d9c4e1a90",
      "name": "Aarav Kumar",
      "moderator": false
    }
  }
}
```

The token is signed with the shared app secret, lives for one hour, and is bound to one room. Teachers and admins get moderator true. Room names are random and never reused.

11.3 Classroom rules

- Students join with microphone and camera off; only moderators can unmute them.
- Lobby is disabled because the JWT already proves identity.
- Chat is on, with a profanity filter applied by the moderator tools; the moderator can disable chat or mute all.
- Recording is optional. When enabled, recording_url is stored on the live class and shown to the class after processing.
- Low bandwidth: the app offers an audio-only toggle and lowers video to 180p automatically after two connection drops.

11.4 Attendance logic

1. First join creates the row with joined_at and zero reconnects.
2. Each later join adds one to reconnects and clears left_at.
3. Leave adds (now minus last join time) to duration_seconds and sets left_at.
4. When the class ends, any open row is closed at the end time.
5. A student counts as present if total duration is at least 10 minutes. The threshold is a setting.

12. Security, Privacy and Compliance

12.1 Technical controls

Area	Control
Transport	HTTPS only with TLS 1.2 or higher; HSTS enabled
Passwords	Salted bcrypt, minimum 8 characters, never logged
Tokens	JWT HS256 or RS256, 12 hour life, secret in environment variable, rotated on compromise
Authorization	Role guard on each route plus server-side class filter for students
Files	Private bucket; only signed URLs of 15 minutes; PDF watermark with student name; file type and size checks; antivirus scan job for uploads
Uploads	Allowed types PDF, JPEG, PNG; magic-byte check, not just extension; random storage keys

Area	Control
Input	Pydantic validation; parameterised queries only through the ORM
Rate limits	Login 5 attempts per 10 minutes per phone; API 120 requests per minute per user
Secrets	Environment variables or a secrets manager; nothing in the repository
Audit	audit_logs for grading, publishing, student creation, role changes, password resets
Backups	Hourly incremental plus daily full database backup; storage versioning; quarterly restore test
Mobile	Token in secure storage; certificate pinning optional; screenshots blocked on PDF viewer where Android allows

12.2 Child data protection

EduBridge stores personal data of minors, so privacy rules apply from day one.

- **India (DPDP Act 2023):** processing a child's data (under 18) needs verifiable consent from a parent or guardian, and the Act prohibits tracking, behavioural monitoring and targeted advertising aimed at children. Plan: the school collects consent at admission and the admin records it per student; the app shows no ads and runs no analytics SDK that profiles students.
- **COPPA (United States):** applies only if the product is offered to children under 13 in the US. It is listed in the original proposal for completeness; confirm applicability before launch.
- **Data minimisation:** store only name, phone, class, roll number and guardian contact. No location, contacts or microphone data are collected outside live classes.
- **Retention:** submissions and attendance are kept for the academic year plus one year, then deleted or anonymised on school request.
- **Parent rights:** a guardian can request a copy or deletion of the child's data through the school admin; the admin tool must support export and delete for one student.
- **Legal review:** these notes are engineering guidance. Have counsel confirm the consent flow and retention policy before the pilot.

13. Non-Functional Requirements

Category	Requirement
Performance	API p95 latency under 300 ms at 500 concurrent users for read endpoints; uploads excluded
App start	Cold start under 3 seconds on a 2 GB RAM Android 8 device
Availability	99.5% monthly target; database and storage failover plan documented
Scalability	One API node serves a 2,000 student school; horizontal scaling behind Nginx for more
Compatibility	Android 8.0 to 15; screens from 5.5 inch phones to 11 inch tablets; admin web on current Chrome, Edge, Firefox, Safari
Network	Resumable uploads; retry with backoff; all lists cached for offline reading
Accessibility	Text scaling, contrast ratio 4.5:1, touch targets at least 48 dp, labels for screen readers
Localisation	English first; string files prepared for Tamil and Hindi
Observability	Structured JSON logs, request ID on every response, health endpoint at /health

14. Technology Stack and Repository Layout

Area	Choice	Reason
Mobile	Flutter 3.x, Dart, Riverpod, Dio, go_router, Hive or Drift for local cache, image_picker, firebase_messaging	One codebase, strong offline and upload support
Admin web	React 19, TypeScript, Vite or Next.js, Tailwind CSS, shadcn/ui, TanStack Query, React Hook Form with Zod	Fast admin tooling with typed forms
Backend	FastAPI, Python 3.12, Pydantic v2, SQLAlchemy 2, Alembic migrations	Async capable, automatic OpenAPI docs

Area	Choice	Reason
Workers	Celery with Redis	Push fan-out, imports, scheduled reminders (Celery beat)
Database	PostgreSQL 16	Constraints, indexes, full-text search
Storage	S3 or Cloudflare R2, local disk for development	Private bucket with presigned URLs
DevOps	Docker, Docker Compose, Nginx, GitHub Actions	Simple single-server start, scalable later

14.1 Repository layout

```
edubridge/
  docker-compose.yml
  README.md
  docs/
  backend/
    app/
      main.py
      config.py
      db.py
      models.py
      schemas.py
      security.py
      deps.py
      storage.py
      notifications.py
      routers/
        auth.py  devices.py  student.py  content.py
        admin.py  assignments.py  live.py  files.py
      workers/
        celery_app.py  tasks.py
    alembic/
    tests/
    Dockerfile
    requirements.txt
  admin-web/
    src/ (pages, components, api, hooks)
    Dockerfile
  mobile/
    lib/ (core, features, shared)
    pubspec.yaml
  deploy/
    nginx.conf
    backup.sh
```

14.2 Environment variables

Variable	Example	Purpose
DATABASE_URL	postgresql+psycopg2://user:pass@db:5432/edubridge	Database connection
SECRET_KEY	long random string	JWT and signed URL secret
ACCESS_TOKEN_MINUTES	720	Token life
STORAGE_BACKEND	local or s3	File storage mode
S3_BUCKET, S3_ENDPOINT, S3_ACCESS_KEY, S3_SECRET_KEY	per provider	Object storage
SIGNED_URL_TTL_SECONDS	900	Signed URL life
REDIS_URL	redis://redis:6379/0	Cache and Celery broker
JITSI_DOMAIN, JITSI_APP_ID, JITSI_APP_SECRET	per server	Live class tokens
FCM_CREDENTIALS_FILE	/run/secrets/fcm.json	Firebase service account
CORS_ORIGINS	https://admin.edubridge.example	Allowed web origins
MAX_UPLOAD_MB	25	Upload limit

15. DevOps and Deployment

15.1 Environments

Environment	Purpose	Data
Local	Developer machine, Docker Compose, SQLite or Postgres	Seed data

Environment	Purpose	Data
Staging	Pilot testing and demos	Anonymised sample data
Production	Live schools	Real data, backups on

15.2 Deployment steps (single server)

1. Provision an Ubuntu 22.04 or 24.04 server with 4 vCPU, 8 GB RAM, 100 GB disk, and a domain with DNS records for api and admin.
2. Install Docker and Docker Compose. Copy the repository and create the .env file from .env.example.
3. Run docker compose up -d to start db, redis, api, worker and nginx.
4. Run alembic upgrade head, then the seed command to create the first Super Admin.
5. Issue TLS certificates with Certbot and enable auto-renewal.
6. Build the admin web and serve it from Nginx.
7. Upload the FCM service account file and set FCM_CREDENTIALS_FILE.
8. Build the Android release (flutter build appbundle), sign it, and upload to Play Console internal testing first.

15.3 CI/CD pipeline (GitHub Actions)

Stage	Action
Lint and type check	ruff and mypy for backend; eslint and tsc for admin; flutter analyze for mobile
Test	pytest with Postgres service container; vitest for admin; flutter test
Build	Docker images tagged with commit SHA; flutter build appbundle
Deploy	On merge to main: push images, SSH to staging, run migrations, smoke test /health. Production deploy needs manual approval

15.4 Backups and monitoring

- pg_dump every night plus WAL archiving or hourly incremental backups; keep 7 daily, 4 weekly, 6 monthly copies.

- Object storage versioning and a lifecycle rule for deleted files.
- Uptime check on /health, error tracking with Sentry, container metrics with Prometheus and Grafana or a hosted equivalent.
- Alerts: API error rate above 2%, disk above 80%, failed backup, push failure rate above 10%.

16. Testing Strategy

Level	Scope	Tools
Unit	Password hashing, status calculation, signed URL signing and expiry, attendance duration maths	pytest
API integration	Every endpoint in section 9 with success and failure cases, including class isolation (Class 4 student requesting Class 10 data must get 404)	pytest with TestClient and a test database
Contract	OpenAPI schema compared in CI so mobile and web clients do not break	schemathesis or openapi-diff
Mobile	Widget tests for login, assignment list and submit; integration test for the full submit flow on an emulator	flutter test, integration_test
Web	Component tests and end-to-end grading flow	vitest, Playwright
Load	500 virtual users on dashboard, assignment list and material endpoints	k6 or Locust
Network	Upload on throttled 2G profile; kill and resume	Android emulator network profiles, Toxiproxy
Security	Auth bypass, IDOR, file type spoofing, expired link reuse, rate limit	OWASP ZAP, manual checks
UAT	One pilot class of 30 students and 3 teachers for two weeks	Feedback form

16.1 Critical end-to-end scenarios

1. Admin creates class, subject, chapter and uploads a PDF; student of that class sees it, student of another class does not.
2. Admin creates an assignment; students receive a push; one student submits three photos; admin grades 22 of 25; student sees marks and remarks.
3. A student submits after the due date with late submission disabled and is refused; with it enabled the submission is marked LATE.
4. Admin starts a live class; students receive a high-priority push; two students join, one drops and reconnects; attendance CSV shows correct duration and reconnect count.
5. A deactivated student cannot log in and an existing token stops working on the next request.
6. A signed file URL stops working after 15 minutes.

17. Roadmap and Effort

17.1 Timeline

The proposal calls this a 12 week plan, but its Gantt dates run from 1 Nov 2026 to about 6 Mar 2027, which is about 18 weeks. The table below uses the Gantt dates, which are the more realistic ones.

Phase	Work package	Start	Length
1 MVP	Requirement validation and database schema	1 Nov 2026	10 days
1 MVP	Backend auth and course API	11 Nov 2026	14 days
1 MVP	Admin web: content and student setup	20 Nov 2026	14 days
1 MVP	Flutter app: login and notes	25 Nov 2026	14 days
1 MVP	Assignment pipeline and file uploads	5 Dec 2026	14 days
1 MVP	Internal alpha testing and hardening	19 Dec 2026	7 days
2 Live and alerts	FCM push pipeline	26 Dec 2026	7 days
2 Live and alerts	Live meeting SDK integration	2 Jan 2027	14 days
2 Live and alerts	Attendance engine and reports	16 Jan 2027	7 days
2 Live and alerts	Beta with pilot school batch	23 Jan 2027	7 days
3 Scale	Teacher portal with scoped access	30 Jan 2027	14 days

Phase	Work package	Start	Length
3 Scale	Parent reporting and offline caching	13 Feb 2027	14 days
3 Scale	Production deployment and store release	27 Feb 2027	7 days

17.2 Deliverables per phase

Phase	Deliverables
1	Backend with auth, classes, students, content, assignments, grading; admin web for these; Flutter app with login, content, offline PDFs, assignment submit and results
2	FCM notifications, live classes with attendance and CSV export, beta release to pilot batch
3	Teacher accounts with scoped permissions, parent weekly digest (WhatsApp Business API or SMS), offline caching hardening, AI doubt solver over syllabus PDFs, Play Store release

17.3 Team assumption

Plan assumes one full stack lead (Harish Kumar D) plus one Flutter developer and one tester or QA part time. With a single developer, add roughly 40% to each phase.

18. Risks and Mitigation

Risk	Severity	Likelihood	Mitigation
Upload failures on 2G and 3G	High	High	Chunked upload with resume; compress images to under 300 KB per page before upload; retry queue stored locally
Low-RAM devices crash	Medium	High	Render PDF pages lazily, release page cache on exit, limit image size, test on 2 GB devices
Live session bandwidth	High	Medium	Audio-only mode, 180p fallback, simulcast on Jitsi

Risk	Severity	Likelihood	Mitigation
Fake submission times	Low	Low	Server timestamp only; device clock ignored
Child data compliance gap	High	Medium	Consent record per student, legal review before pilot, no analytics SDKs
FCM delivery delays on aggressive battery savers	Medium	Medium	In-app notification list, in-app polling on app open, guidance screen for battery settings
Storage and video cost growth	Medium	Medium	Compress video, lifecycle rules, cap video length, use R2 to avoid egress fees
Single developer bottleneck	Medium	Medium	Strict phase scope, OpenAPI contract first so web and mobile can progress in parallel
Zoom or Jitsi provider change	Low	Low	Provider field and token logic isolated in one module

19. Acceptance Criteria and Definition of Done

- ☐ Android app runs smoothly on Android 8.0 to 15 and on 5.5 inch to 11 inch screens.
- ☐ 95th percentile API latency is below 300 ms with 500 concurrent users on read endpoints.
- ☐ A homework capture made offline is kept locally and uploaded automatically once the network returns; no data loss.
- ☐ Student of one class cannot read any content, assignment or live room of another class (verified by automated tests).
- ☐ Signed file URLs expire after 15 minutes and cannot be reused.
- ☐ Full assignment loop works: create, push, submit, grade, notify, answer key.
- ☐ Live class attendance matches a manual check for a test session of at least 10 students.
- ☐ Hourly incremental backups run and a restore test succeeds.
- ☐ All endpoints in section 9 are covered by automated tests and appear in the generated OpenAPI document.
- ☐ Security checklist in section 12 is signed off and the consent process is approved.

Appendix A. Error Code Catalogue

Code	HTTP	Meaning
INVALID_CREDENTIALS	401	Wrong phone or password
ACCOUNT_DISABLED	401	User is deactivated
TOKEN_EXPIRED	401	Access token expired; log in again
FORBIDDEN	403	Role not allowed for this endpoint
NOT_YOUR_CLASS	403	Live class or resource belongs to another class
LINK_EXPIRED	403	Signed file link expired or invalid
NOT_FOUND	404	Resource missing or hidden
PHONE_EXISTS	409	Phone number already registered
ROLL_EXISTS	409	Roll number already used in the class
CLASS_EXISTS	409	Class, section and year already exist
LATE_NOT_ALLOWED	400	Due date passed and late submission is off
ALREADY_GRADED	400	Submission is final
CLASS_NOT_STARTED	400	Live class cannot be joined yet
CLASS_ENDED	400	Live class is over
FILE_TOO_LARGE	413	Upload exceeds limit
UNSUPPORTED_FILE_TYPE	422	File type not allowed
VALIDATION_ERROR	422	Body or query invalid
RATE_LIMITED	429	Too many requests
INTERNAL_ERROR	500	Unexpected failure

Appendix B. Sample Student Import CSV

```
full_name,phone_number,grade_level,section,roll_number,guardian_name,guardian_phone
Aarav Kumar,9876543210,8,A,07,Suresh Kumar,9876500000
Diya Sharma,9876543211,8,A,08,Meena Sharma,9876500001
Kabir Singh,9876543212,9,B,12,Harpreet Singh,9876500002
```

Appendix C. Changes from Proposal v1.0

Item	Proposal v1.0	This document	Reason
Role name	ER diagram says ADMIN, DDL says SUPER_ADMIN	SUPER_ADMIN everywhere	One consistent enum
Submission files	One submission_file_url	file_keys list	Students upload several photos
File fields	file_url columns	file_key storage keys	Signed URL generation; no public links
Answer key	Single URL	URL plus answer_key_published flag	Hidden until staff publish
Chapters	No release date	release_at	Scheduled topic release requirement
Attendance	Join, leave, duration	Plus reconnects	Connection drop tracking requirement
New tables	None	device_tokens, audit_logs	FCM tokens and audit trail
Timeline	Titled 12 weeks	About 18 weeks per Gantt dates	Dates did not match title
Live provider	Zoom or Jitsi, unspecified	Self-hosted Jitsi with JWT recommended	Public Jitsi cannot restrict rooms
Compliance	COPPA and DPDP together	DPDP primary, COPPA only if US children	COPPA applies to US users under 13

Appendix D. Glossary

Term	Meaning
RBAC	Role-based access control

Term	Meaning
JWT	JSON Web Token, signed token carrying the user identity
FCM	Firebase Cloud Messaging, Google's push service
Signed URL	A link with a signature and expiry that grants temporary access to one file
DPDP	India's Digital Personal Data Protection Act, 2023
MVP	Minimum viable product, the Phase 1 release
IDOR	Insecure direct object reference, accessing data by guessing an ID

Appendix E. Decisions to Confirm Before Build

1. Live class provider: self-hosted Jitsi, 8x8 JaaS or Zoom.
2. Storage: AWS S3 or Cloudflare R2, and the hosting region (India region recommended for latency).
3. Login method for Classes 3 to 5: phone and password, or roll number and school code.
4. Parent notifications channel for Phase 3: WhatsApp Business API or SMS.
5. Whether teachers need separate logins in Phase 1 or only the Super Admin is used.
6. Pilot school, class strength and go-live date.